

Contractive metrics for scalar conservation laws.

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Title in English

Scalar conservation laws, a probabilistic approach.

Título en español

Leyes de conservación escalar, un enfoque probabilístico.

Abstract: We study a certain class of partial differential equation known as scalar conservation law, we focus on certain probabilistic issues that naturally arise when studying existence and uniqueness of a solution for the given equation. More explicitly we take as initial data a probability cumulative distribution function and we prove that at each time, the solution is always again a probability cumulative distribution function. We introduce a metric called the Wasserstein metric, also known as Monge Kantorovich metric, and study metric properties of the solution respect to the initial data.

Resumen: En este trabajo se estudia un tipo de ecuación en derivadas parciales conocido como ley de conservación escalar, pero desde el punto de vista de la teoría de la probabilidad. Se estudia como, al tomar como dato inicial una función de distribución acumulativa de probabilidad, la solución de la ecuación diferencial es nuevamente una función de distribución acumulativa de probabilidad para cualquier instante de tiempo. Definimos una métrica conocida como la métrica de Wasserstein o métrica de Monge Kantorovich y estudiamos como se comporta la solución respecto al dato inicial.

Keywords: Scalar conservation laws, Probability, Wasserstein Metric.

Palabras clave: Leyes de conservación escalar, Probabilidad, Métrica de Wasserstein.

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Introduction

The aim of this paper is to study, almost from the very basics, the continuous dependence among solutions and initial data of a Cauchy problem related to an *scalar conservation law*.

We begin by setting a class of functions $\mathcal{U}$, which plays a central roll in different branches of knowledge such as statistics. For this class of functions $\mathcal{U}$ we first derive a kind of “fundamental theorem of calculus” which shows a relation between differentiation, and how to recover functions from its derivatives via an accumulation process. This class of functions is also studied from a very abstract viewpoint, known in category theory as the class of *left adjoint functions*, which turns out to be a very important tool for computations.

About the equation itself in the year of 1969 the soviet mathematician *S. Kružkov* showed existence and uniqueness of solutions. The reason for considering this class of functions $\mathcal{U}$, is that it remains invariant for the conservation law; if the initial data is a function in $\mathcal{U}$ then so does the solution at every time t .

We then proceed to study certain continuous dependence between the initial data and solutions; for this we introduce the *Wasserstein metric*. We point out that among the many different ways to provide a metric structure on a set of probability measures, this one has shown to be useful, not only in the field of statistics but also in *transportation theory*. In our case we will make some measurements with it, since derivatives of functions of $\mathcal{U}$ can be regarded as probability measures.

With all the machinery ready for action, we construct an approximation for the solution of the scalar conservation law, and prove many different convergence results, that pave the way down towards the desired result, the Wasserstein distance between solutions is a non increasing function of time, and this yields a continuous dependence between solutions and initial data.

We end up this article by showing how to apply the whole theory developed in this article to actually find explicit solutions to equations such as the *Burger's equation* and the *linear advection equation* as long as the initial data is a function in $\mathcal{U}$.

We tried to be as self contained as possible, trying to state all the results needed and, for sake of consistency, we put references wherever needed for the detailed proofs that lie beyond the scope of this paper.

CHAPTER 1

Probability

1.1 Probability.

1.1.1 Some basic probability

In this section we settle some terminology and useful notation that will be used throughout the paper.

Consider $\mathbb{R}$ or the open interval $(0, 1) \subset \mathbb{R}$. We say that μ is a probability measure on $\mathbb{R}$ or $(0, 1)$, whenever for any of these spaces, endowed with their Borel σ -algebra, μ is a Borel measure such that $\mu(\mathbb{R}) = 1$ or $\mu((0, 1)) = 1$. Unless stated explicitly we shall denote $(0, 1)$ for the probability space $(0, 1)$ equipped with its Borel σ -algebra and dw will stand for the *Lebesgue* measure on it.

Given any probability measure μ on $\mathbb{R}$ we define its cumulative distribution function by means of

$$F_\mu(x) = \mu((-\infty, x]) \text{ for every } x \in \mathbb{R}.$$

This distribution function satisfies some interesting properties as the following lemma shows.

Lemma 1.1.1 *Let μ be a probability measure on $\mathbb{R}$ and let F_μ be its cumulative distribution function. Then the following holds.*

1. If $x \leq y$ then $F_\mu(x) \leq F_\mu(y)$,
2. $\lim_{x \rightarrow -\infty} F_\mu(x) = 0$,
3. $\lim_{x \rightarrow \infty} F_\mu(x) = 1$,
4. $\lim_{h \rightarrow 0^+} F_\mu(x + h) = F_\mu(x)$.

Proof: In this proof we use some properties of measures that can be found in [3].

1. Let $x \leq y$ then $(-\infty, x] \subset (-\infty, y]$ and thus $\mu((-\infty, x]) \leq \mu((-\infty, y])$ as claimed.
2. Let $\{x_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$ be any decreasing sequence such that:

$$\lim_{n \rightarrow \infty} x_n = -\infty.$$

Then family of sets $\{(-\infty, x_n]\}_n$ is such that

$$\bigcap_{n=1}^{\infty} (-\infty, x_n] = \emptyset,$$

and for $n \leq m$

$$(-\infty, x_n] \subset (-\infty, x_m].$$

Then since μ is a probability measure; $\mu(\mathbb{R}) = 1$, we have that

$$0 = \mu(\emptyset) = \mu\left(\bigcap_{n=1}^{\infty} (-\infty, x_n]\right) = \lim_{n \rightarrow \infty} \mu((-\infty, x_n]).$$

And we are done since the sequence is arbitrary.

3. The proof is similar to the one above, we just use the property that for any increasing family of sets $\{E_n\}_{n \in \mathbb{N}}$

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \mu(E_n).$$

This property holds regardless μ is a probability measure on $\mathbb{R}$ or just a measure in general.

4. For this proof, let $x \in \mathbb{R}$ and choose a sequence $\{h_n\}_{n \in \mathbb{N}}$ such that $h_n \rightarrow 0^+$ in a monotone fashion. We then have that the family of sets $\{(-\infty, x + h_n]\}_n$ is a decreasing family of sets such that

$$\bigcap_{n=1}^{\infty} (-\infty, x + h_n] = (-\infty, x].$$

Again using the fact that μ is a probability measure on $\mathbb{R}$ we can pass the limits to obtain

$$\mu((-\infty, x]) = \lim_{n \rightarrow \infty} \mu\left(\bigcap_{n=1}^{\infty} (-\infty, x + h_n]\right),$$

and since $\{h_n\}_n$ was arbitrary but positive, the proof follows. ■

The following notation will be used throughout the paper.

Definition 1.1.1 We say that a function

$$F : \mathbb{R} \rightarrow [0, 1]$$

belongs to $\mathcal{U}$ whenever the following conditions hold:

1. If $x \leq y$ then $F(x) \leq F(y)$,
2. $\lim_{x \rightarrow -\infty} F(x) = 0$,
3. $\lim_{x \rightarrow \infty} F(x) = 1$,
4. $\lim_{h \rightarrow 0^+} F(x+h) = F(x)$.

As should be clear, this class of functions contains the cumulative distribution function of any given probability measure μ on $\mathbb{R}$. This gives rise to the question whether or not can be, somehow, recover a probability measure μ from its distribution function F_μ , turning $\mathcal{U}$ into precisely the class of such functions. Towards this question we have the following theorem which aims to prove that this is exactly the case.

The next theorem can be regarded as “fundamental theorem of calculus” for it shows that accumulation is the inverse operation to that of differentiation. Although there exists a more general setting under which the theorem can be proved, see [9], we are going to restrict ourselves to the case of probability measures.

For the proof of the following theorem recall that

Definition 1.1.2 *Let*

$$u : \mathbb{R} \rightarrow \mathbb{R}.$$

We say u is of locally bounded variation if for any $a < b$ and any partition of the interval $a < x_1 < \dots < x_{n+1} = b$ the following

$$\sup \sum_{j=1}^n |u(x_{j+1}) - u(x_j)|$$

where the supremum is extended over all partitions of $(a, b]$ exists.

This supremum is usually denoted by:

$$V_a^b u.$$

Remark 1 *Every function $u \in \mathbb{U}$ is a function of locally bounded variation indeed we have that*

$$V_a^b u = u(b) - u(a^+).$$

This is easily verified since

$$\sum_{j=1}^n |u(x_{j+1}) - u(x_j)| = \sum_{j=1}^n u(x_{j+1}) - u(x_j) = u(b) - u(x_1)$$

and thus the claim follows.

In order to state the following theorem, we must understand first what we mean by “differentiation” of functions $u \in \mathcal{U}$. It should be clear that we are not allowed to take

classical derivatives such as the ones we all are taught in a first course of calculus, since functions in $\mathcal{U}$ may well have infinite points of discontinuity, and no derivatives can be taken at those points, however we will show a way to overcome such obstacle.

Suppose we are given the following data: we have a function $u \in \mathcal{U}$ and a function $\varphi \in \mathcal{C}_c^1(\mathbb{R}, \mathbb{R})$. Assume also that u has classical derivative u_x , then we find by integrating by parts that:

$$\int_{\mathbb{R}} u_x \varphi(x) dx = - \int_{\mathbb{R}} u \varphi'(x) dx$$

but notice that the right hand side does not involve information about the differentiability of the function u . This motivates the following definition.

Definition 1.1.3 *Let $u \in \mathcal{U}$. We say that a measurable real valued function g is a weak derivative of u if the following formula holds true for all $\varphi \in \mathcal{C}_c^1(\mathbb{R}, \mathbb{R})$.*

$$\int_{\mathbb{R}} g(x) \varphi(x) dx = - \int_{\mathbb{R}} u \varphi'(x) dx.$$

In case such a function g exists we denote it by u_x .

We now proceed to state and proof:

Theorem 1.1.1 *The weak derivative of any $u \in \mathcal{U}$ is a probability measure on $\mathbb{R}$*

Proof:

1. We first prove the following fact about arbitrary functions $u \in \mathcal{U}$,

$$\forall V \subset \subset \mathbb{R}, \sup \left\{ \int_V u \varphi' dx \mid \varphi \in \mathcal{C}_c^1(V, \mathbb{R}), \|\varphi\|_{\infty} \leq 1 \right\} \leq 1.$$

Let $\varepsilon > 0$ and put

$$u_{\varepsilon} = \rho_{\varepsilon} \star u$$

The usual smooth approximation of u by means of an approximate identity ρ_{ε} . We consider the case V is a connected set i.e, $V = (a, b)$ an open interval. Choose t_i , $i = 1, \dots, m+1$ such that $a + \varepsilon < t_1 < \dots < t_{m+1} < b - \varepsilon$.

$$\begin{aligned} \sum_{j=1}^m |u_{\varepsilon}(t_{j+1}) - u_{\varepsilon}(t_j)| &= \sum_{j=1}^m \left| \int_{-\varepsilon}^{\varepsilon} \rho_{\varepsilon}(s) u(t_{j+1} - s) - u(t_j - s) ds \right| \\ &\leq \sum_{j=1}^m \int_{-\varepsilon}^{\varepsilon} \rho_{\varepsilon}(s) |u(t_{j+1} - s) - u(t_j - s)| ds \\ &= \int_{-\varepsilon}^{\varepsilon} \rho_{\varepsilon}(s) \sum_{j=1}^m |u(t_{j+1} - s) - u(t_j - s)| ds \\ &\leq \int_{-\varepsilon}^{\varepsilon} \rho_{\varepsilon}(s) ds \\ &= 1 \end{aligned}$$

By the mean value theorem it follows that:

$$\sum_{j=1}^m |u_\varepsilon(t_{j+1}) - u_\varepsilon(t_j)| = \sum_{j=1}^m |u'_\varepsilon(t'_j)|(t_{j+1} - t_j) \text{ for some } t_j < t'_j < t_{j+1}, j = 1, \dots, m,$$

and thus

$$\int_{a+\varepsilon}^{b-\varepsilon} |u'_\varepsilon| dx = \sup \left\{ \sum_{j=1}^m |u_\varepsilon(t_{j+1}) - u_\varepsilon(t_j)| t_j \text{ partition of } (a+\varepsilon, b-\varepsilon) \right\} \leq 1.$$

Let $\varphi \in \mathcal{C}_c^1(V, \mathbb{R})$ with $\|\varphi\| \leq 1$

$$\int_a^b u_\varepsilon \varphi' dx = - \int_a^b u'_\varepsilon \varphi dx \leq \int_{a+\varepsilon}^{b-\varepsilon} |u'_\varepsilon| dx \leq 1$$

as $\varepsilon \rightarrow 0$ we get

$$\int_a^b u \varphi' dx \leq 1$$

proving our assertion.

With this fact about functions in $\mathcal{U}$ we proceed to prove the theorem.

2. Define the following linear functional

$$L : \mathcal{C}_c^1(U, \mathbb{R}) \rightarrow \mathbb{R}$$

$$L(\varphi) = - \int_U u \varphi' dx.$$

L is clearly a lineal functional, and our previous assertion allows us to define the following

$$C(V) = \sup \{ L(\varphi) | \varphi \in \mathcal{C}_c^1(V, \mathbb{R}), \|\varphi\|_\infty \leq 1 \} \leq 1$$

for any $V \subset\subset U$, and thus

$$|L(\varphi)| \leq C(V) \|\varphi\|_\infty.$$

Now, for any $\varphi \in \mathcal{C}_c(U, \mathbb{R})$ fix a compact set $K \subset U$, and an open set V such that $\text{supp}(\varphi) \subset K \subset V \subset\subset U$. Then there always exists a sequence $\{\varphi_k\}_k \subset \mathcal{C}_c^1(V, \mathbb{R})$ such that $\varphi_k \rightarrow \varphi$ uniformly on V , and therefore in U by extending the functions φ_k by 0 outside V . We use this sequence to extend the linear functional L to $\bar{L}$ by means of the following formula:

$$\bar{L}(\varphi) = \lim_{k \rightarrow \infty} L(\varphi_k).$$

This limit always exists and it is well defined, in the sense that it is independent of the choice of the sequence $\{\varphi_k\}_k$ converging to φ , since L is bounded and $\mathcal{C}_c^1(U, \mathbb{R})$ is dense in $\mathcal{C}_c(U, \mathbb{R})$.

This unique extension

$$\bar{L} : \mathcal{C}_c(U, \mathbb{R}) \rightarrow \mathbb{R}$$

also satisfies that

$$\sup\{\bar{L}(\varphi) | \varphi \in \mathcal{C}_c(U, \mathbb{R}), \|\varphi\|_\infty \leq 1, \text{supp}(\varphi) \subset K\} \leq 1$$

for each $K \subset U$.

We now make use of the well known *Riesz representation theorem*, see for instance [9], to grant the existence a *Radon* measure μ such that

$$\int_U u\varphi' dx = - \int_U \varphi d\mu$$

for all $\varphi \in \mathcal{C}_c^1(U, \mathbb{R})$.

3. Now we ought to check that μ is actually a probability measure, in symbols $\mu(\mathbb{R}) = 1$.

From the Riesz representation theorem, it follows that for every $a, b \in \mathbb{R}$

$$\mu((a, b)) = \sup \left\{ \int_a^b u\varphi' dx \mid \varphi \in \mathcal{C}_c^1((a, b), \mathbb{R}), \|\varphi\|_\infty \leq 1 \right\}$$

We will show that $u(b) - u(a^+) \leq \mu((a, b))$.

Choose $a < c < d < b$ and $\varphi \in \mathcal{C}_c^1((c, d), \mathbb{R})$ with $\|\varphi\|_\infty \leq 1$. For each adequately small $\varepsilon > 0$, we have that

$$\begin{aligned} \int_c^d u'_\varepsilon \varphi dx &= - \int_c^d u_\varepsilon \varphi' dx \\ &= - \int_c^d (\rho_\varepsilon \star u) \varphi' dx \\ &= - \int_a^b u(\rho_\varepsilon \star \varphi)' dx \\ &\leq \mu((a, b)) \end{aligned}$$

Thus

$$\int_c^d |u'_\varepsilon| dx \leq \mu((a, b)).$$

Finally by choosing a partition $a < t_1 < \dots < t_{m+1} < b$,

$$\begin{aligned} \sum_{j=1}^m |u(t_{j+1}) - u(t_j)| &= \lim_{\varepsilon \rightarrow 0} \sum_{j=1}^m |u_\varepsilon(t_{j+1}) - u_\varepsilon(t_j)| \\ &\leq \limsup_{\varepsilon \rightarrow 0} \int_a^b |u'_\varepsilon| dx \\ &= \mu((a, b)) \end{aligned}$$

and so $u(b) - u(a^+) \leq \mu((a, b))$ as claimed.

$$1 = \lim_{b \rightarrow \infty} u(b) - \lim_{a \rightarrow \infty} u(a) \leq \mu(\mathbb{R}) \leq 1,$$

and this completes the proof.

■

Again we would like to point out that the crucial point in this proof is the fact that any $v \in \mathcal{U}$ is a function on *Bounded variation*. A good introduction of functions of bounded variation can be found in [11] and for a more advanced treatment we refer to [9].

One more comment about this theorem. We begin with a probability measure μ on $\mathbb{R}$, then we took its cumulative distribution function F_μ and then we find its weak derivative $(F_\mu)_x$. The theorem above not only shows us that $(F_\mu)_x$ is probability measure, but, putting step 1 and step 3 together, we actually find that

$$\begin{aligned} (F_\mu)_x((-\infty, y]) &= \sup \left\{ \int_{-\infty}^y F_\mu \varphi' dx \mid \varphi \in \mathcal{C}_c^1((-\infty, y], \mathbb{R}) \right\} \\ &= F_\mu(y) - F_\mu(-\infty) = F_\mu(y) = \mu((-\infty, y]). \end{aligned}$$

Thus we did not recover any but the original probability measure μ .

1.1.2 Convergence of probability measures.

Theorem 1.1.2 *Let μ, μ_n with $n \in \mathbb{N}$ be probability measures on $\mathbb{R}$. The following statements are equivalent:*

1. $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f d\mu_n = \int_{\mathbb{R}} f d\mu$ for all $f \in \mathcal{C}_c(\mathbb{R}, \mathbb{R})$.
2. $\limsup_{n \rightarrow \infty} \mu_n(K) \leq \mu(K)$ for all $K \subset \mathbb{R}$ compact and $\mu(U) \leq \liminf_{n \rightarrow \infty} \mu_n(U)$ for each $U \subset \mathbb{R}$ open.
3. $\lim_{n \rightarrow \infty} \mu_n(B) = \mu(B)$ for each Borel set $B \subset \mathbb{R}$ with $\mu(\partial(B)) = 0$.

Proof: We refer to [9].

■

With this theorem at hand we define the *weak convergence* of probability measures.

Definition 1.1.4 *Let μ, μ_n with $n \in \mathbb{N}$ be probability measures on $\mathbb{R}$. If any, and thus all of the conditions of the previous theorem is satisfied, we say that μ_n converges weakly to the measure μ . We write*

$$\mu_n \rightharpoonup \mu$$

in this case.

There is another useful criteria to determine weak convergence of probability measures. Let $F_\mu(x) = \mu((-\infty, x])$ be the cumulative distribution function of the probability measure μ , then we have:

Lemma 1.1.2 *Let μ, μ_n with $n \in \mathbb{N}$ be probability measures on $\mathbb{R}$. Then $\mu_n \rightharpoonup \mu$ if and only if $F_{\mu_n} \rightarrow F_\mu$ at each point x where F_μ is continuous.*

Remark 2 *The set $D = \{x \in \mathbb{R} \mid \mu(x) = 0\}$ is dense in $\mathbb{R}$ since it is the set of points of continuity of F_μ , an increasing real valued function.*

Proof:

1. Suppose first that $F_{\mu_n} \rightarrow F_\mu$ on D , as above. Choose $\varepsilon > 0$, $\varphi \in \mathcal{C}_c(\mathbb{R}, \mathbb{R})$ and $K > 0$ such that $\mu(-K, K) > 1 - \varepsilon$. We can choose $a \leq -K$ and $b > K$ such that $a, b \in D$. Since D is dense.

Let $a = x_0 < x_1 < x_2 < \dots < x_m = b$ a partition such that $x_0, \dots, x_m \in D$ and

$$\sup_{x_i \leq x \leq x_{i+1}} |\varphi(x) - \varphi(x_i)| < \varepsilon$$

for all $i = 0, 1, \dots, m-1$. Such partition can always be chosen because φ is uniformly continuous on $[a, b]$.

$$\begin{aligned} \left| \int_{\mathbb{R}} \varphi d\mu_n - \int_{\mathbb{R}} \varphi d\mu \right| &= \left| \int_{-\infty}^a \varphi d\mu_n - \int_{-\infty}^a \varphi d\mu \right. \\ &\quad \left. + \sum_{i=1}^{m-1} \left(\int_{x_i}^{x_{i+1}} \varphi d\mu_n - \int_{x_i}^{x_{i+1}} \varphi d\mu \right) + \int_b^{\infty} \varphi d\mu_n - \int_b^{\infty} \varphi d\mu \right| \\ &\leq \|\varphi\|_{\infty} (\mu_n((-\infty, a]) + \mu((-\infty, a]) + \\ &\quad \sum_{i=0}^{m-1} |\mu_n((x_i, x_{i+1}]) - \mu((x_i, x_{i+1}])| + \mu_n((b, \infty)) + \mu((b, \infty))) \end{aligned}$$

Letting $n \rightarrow \infty$ we get that

$$\limsup_{n \rightarrow \infty} \left| \int_{\mathbb{R}} \varphi d\mu_n - \int_{\mathbb{R}} \varphi d\mu \right| \leq \|\varphi\|_{\infty} 2(1 - \mu((a, b])) \leq \|\varphi\|_{\infty} 2\varepsilon,$$

and the result follows since ε is arbitrary.

2. Let $\varepsilon > 0$, $a, b \in D$ such that $a < b$. Let $\varphi \in \mathcal{C}_c(\mathbb{R}, \mathbb{R})$ such that $\varphi = 1$ on $[a, b]$, $\text{supp}(\varphi) \subset (a - \varepsilon, b + \varepsilon)$ and $0 \leq \varphi \leq 1$. Since $\mu_n \rightarrow \mu$ it follows that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} \varphi d\mu_n = \int_{\mathbb{R}} \varphi d\mu,$$

but $\mu_n((a, b]) \leq \int_{\mathbb{R}} \varphi d\mu_n$ and $\int_{\mathbb{R}} \varphi d\mu \leq \mu((a - \varepsilon, b + \varepsilon])$, thus

$$\limsup_{n \rightarrow \infty} \mu_n((a, b]) \leq \mu((a - \varepsilon, b + \varepsilon]).$$

Finally by letting $\varepsilon \rightarrow 0$ we get

$$\limsup_{n \rightarrow \infty} \mu_n((a, b]) \leq \mu((a, b]).$$

A similar argument with $\phi \in \mathcal{C}_c(\mathbb{R}, \mathbb{R})$ such that $\text{supp}(\phi) \subset (a, b)$, $0 \leq \phi \leq 1$ and $\phi = 1$ on $[a - \varepsilon, b + \varepsilon]$, yields

$$\liminf_{n \rightarrow \infty} \mu_n((a, b]) \geq \mu((a, b]).$$

Thus

$$\lim_{n \rightarrow \infty} \mu_n(a, b] = \mu(a, b]$$

for all $a, b \in D$ which is dense. ■

We end this section with a theorem about compactness of probability measures. The theorem is due to *Helly* and is usually known as *Helly's selection theorem*.

Theorem 1.1.3 *Let $\mathcal{A}$ be an infinite collection of probability measures on $\mathbb{R}$. Then there exist a sequence $\{\mu_n\}_n \subset \mathcal{A}$ and a probability measure μ such that $\mu_n \rightharpoonup \mu$.*

Proof: The proof is beautifully written in [2]. ■

For a more complete and detailed exposition about weak convergence of probability measures we recommend the book of [2].

1.1.3 Product of measures.

In this section we shall merely mention some concepts that will be used later. For a fully detailed and fine exposition we recommend [3].

Given a probability measure π on $\mathbb{R}^2$, we say that π has μ_1 and μ_2 as its marginals provided that

$$\begin{aligned}\mu_1(A) &= \pi(A \times \mathbb{R}) \text{ for all } A \subset \mathbb{R} \text{ measurable and,} \\ \mu_2(B) &= \pi(\mathbb{R} \times B) \text{ for all } B \subset \mathbb{R} \text{ measurable.}\end{aligned}$$

Given two probability measures μ_1, μ_2 we define

$$\Pi(\mu_1, \mu_2) = \{\pi | \pi \text{ is a probability measure on } \mathbb{R}^2 \text{ with marginals } \mu_1 \text{ and } \mu_2\}$$

This set is never empty since the product measure $\mu_1 \times \mu_2 \in \Pi(\mu_1, \mu_2)$.

1.1.4 Random variables.

Definition 1.1.5 *A random variable X is a measurable real function on a probability space.*

Random variables give rise to probability measures on the real line $\mathbb{R}$ in a rather natural way.

Definition 1.1.6 *Given any random variable $X : (0, 1) \rightarrow \mathbb{R}$, and given Borel set $B \subset \mathbb{R}$, let*

$$\mu_X(B) = \int_0^1 1_{\{w \in (0, 1) | X(w) \in B\}}(w) dw.$$

It is easy to check that μ_X is actually a probability measure on $\mathbb{R}$, and such measure is called the law of the random variable X . Following [4] given any random variable X on the probability space $(\mathbb{R}, \mu)$, we can define its probability cumulative function, or simply distribution function, as

$$F_X(y) = \mu(X^{-1}((-\infty, y])),$$

which happens to be exactly the cumulative distribution function of the measure μ_X .

In general any of such distribution functions satisfies the following properties by lemma 1.1.1:

1. If $x \leq y$ then $F_X(x) \leq F_X(y)$,
2. $\lim_{x \rightarrow -\infty} F_X(x) = 0$,
3. $\lim_{x \rightarrow \infty} F_X(x) = 1$,
4. $\lim_{h \rightarrow 0^+} F_X(x+h) = F_X(x)$.

Related to the measure μ_X we have the following “change of variable formula”.

Lemma 1.1.3 *Let $X : (0, 1) \rightarrow \mathbb{R}$ be a random variable, and let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function. Then we have:*

$$\int_0^1 \varphi(X(w))dw = \int_{\mathbb{R}} \varphi(x)d\mu_X(x).$$

Proof: Following [3], Recall that a *simple* function is a measurable function whose range is finite. Following [3], the result will follow by proving it first for simple functions.

Let $g : (\mathbb{R}, \mu_X) \rightarrow \mathbb{R}$ be any simple function whose range is the finite set $\{b_i\}_{i=1}^n$,

$$g(x) = \sum_{i=1}^n b_i 1_{\{x \in \mathbb{R} | g(x) = b_i\}}(x),$$

the integral of g can be regarded as the integral of the simple function $g \circ X$

$$\int_{\mathbb{R}} g d\mu_X = \sum_{i=1}^n b_i \int_0^1 1_{\{w \in (0,1) | g(X(w)) = b_i\}}(w)dw = \int_0^1 g \circ X(w)dw.$$

Assume now that $\varphi \geq 0$. Then by [3] there always exists a sequence of simple functions g_m such that,

- $0 \leq g_m(x) \leq g_{m+1}(x)$ for all $x \in \mathbb{R}$
- $\varphi(x) = \lim g_m(x)$ for all $x \in \mathbb{R}$

Using the well known monotone convergence theorem we get:

$$\int_{\mathbb{R}} \varphi(x)d\mu_X(x) = \lim_m \int_{\mathbb{R}} g_m(x)d\mu_X(x) = \lim_m \int_0^1 g_m(X(w))dw = \int_0^1 \varphi(X(w))dw.$$

Finally apply this argument for the positive and negative parts of φ . ■

We end this section with the following terminology. A probability measure μ on $\mathbb{R}$ has finite moment of order p if

$$\int_{\mathbb{R}} |x|^p d\mu < \infty.$$

Definition 1.1.7 $v \in \mathcal{U}_p$ if and only if $v \in \mathcal{U}$ and v_x has finite moment of order p .

Wasserstein metrics

2.1 Wasserstein metrics

In this section we introduce the *Wasserstein metric* along with some of its properties. This metric will be used later on when dealing with solutions of a certain class of partial differential equations.

Definition 2.1.1 *Let $p \geq 1$ and μ_1, μ_2 be probability measures on $\mathbb{R}$*

$$W_p(\mu_1, \mu_2) = \inf_{\pi \in \Pi(\mu_1, \mu_2)} \left(\int_{\mathbb{R}^2} |x - y|^p d\pi(x, y) \right)^{1/p}$$

The Wasserstein metric is actually a minimization problem and it is a non-trivial fact to show that this problem always admits a solution. In the literature it is also known under different names such as the *Monge-Kantorovich distance*, or taking $p = 1$ it is sometimes called the *Kantorovich-Rubinstein distance* and when $p = 2$ it is also named as the *quadratic Wasserstein distance*. We will simply call them the *Wasserstein distance*. This minimization problem arose from the study of *optimal transport mass* back in the year of 1781 by the French mathematician *Monge*, the problem was then retaken in the offspring of world war II almost 150 years after Monge, by the Soviet economist *Kantorovich*.

More recently the French mathematician *C. Villani*, awarded the *Fields medal* in 2010, wrote a book called *topics in optimal transportation* [13] concerned entirely about this problem. We suggest the reader to read this book carefully for it is an excellent and complete book on this subject, In this book a proof about the existence of a solution that minimizes the Wasserstein metric can be found .

Our task now is to show an equivalent way of defining the Wasserstein metric in terms of random variables, and then show how this number can be reckoned, at least theoretically. For this we introduce the concept of a generalized inverse or *left adjoint* function, it is both remarkable and surprising that the following construction is an ubiquitous and central idea through mathematics, many deep theorems such as *Galois' theorem* for field extension and subgroups lies upon this notion.

Definition 2.1.2 Let $u \in \mathcal{U}$, we define its left adjoint function or generalized inverse function by

$$u^{-1} : (0, 1) \rightarrow \mathbb{R}$$

$$u^{-1}(w) = \inf\{x \in \mathbb{R} | u(x) \geq w\}.$$

Maybe the most important easy to check properties about this function are summarized in the following lemma:

Lemma 2.1.1 Let $u \in \mathcal{U}$ and its generalized inverse u^{-1} . Then the following properties hold.

1. If $w_1 \leq w_2$ then $u^{-1}(w_1) \leq u^{-1}(w_2)$.
2. $u^{-1}(w) \leq x \iff w \leq u(x)$.
3. u^{-1} is continuous from the right.

Proof: Proofs are straight forward from the definition. ■

In our case, u^{-1} yields another interesting and useful fact.

Lemma 2.1.2 The law of the random variable u^{-1} is u_x .

Proof: Following [3] it suffices to check that for any $x \in \mathbb{R}$

$$\mu_{u^{-1}}((-\infty, x]) = u_x((-\infty, x]) = u(x),$$

keep in mind that $\mu_{u^{-1}}$ is the law of u^{-1} .

$$\begin{aligned} \mu_{u^{-1}}((-\infty, x]) &= \int_0^1 1_{\{w \in (0,1) | u^{-1}(w) \leq x\}}(w) dw \\ &= \int_0^1 1_{\{w \in (0,1) | w \leq u(x)\}}(w) dw \\ &= u(x) \end{aligned}$$

Moreover the cumulative distribution function is u . ■

This lemma shows a very important feature in probability theory for it shows that the probability space $(0, 1)$ is universal in the sense that it is always possible to find a random variable X on $(0, 1)$ with cumulative distribution function a prescribed function in $\mathcal{U}$ or, up to differentiation, a random variable whose law is any given probability measure on $\mathbb{R}$.

We now proceed to derive some useful consequences of this fine lemma, but first lets take the probabilistic view point of what has been done so far.

In the given definition of the Wasserstein metric, one can replace all the measure theoretical language by the language of random variables. Instead of taking probability measures μ_1 and μ_2 on $\mathbb{R}$, we can replace them by random variables on the space $(0, 1)$ with respective laws μ_1 and μ_2 , and instead of measures on the product space, we can

think about *couplings* between them. By doing so we will show how to use the generalized inverse to actually reckon the distance W_p .

Remark 3 *Given $u, \bar{u} \in \mathcal{U}$, then the probability measures on $\mathbb{R}$ $u_x, \bar{u}_x$ satisfy*

$$W_p(u_x, \bar{u}_x) = \left(\int_0^1 |u^{-1}(w) - \bar{u}^{-1}(w)|^p dw \right)^{1/p}$$

The proof of this fact lies beyond the scope of this paper, we suggest the interested reader to take a look at the book of [13] for a detailed and more general description of the situation.

We will now show that W_p actually defines a metric on $\mathcal{P}_p$.

Lemma 2.1.3 *Let $p \geq 1$, W_p is actually a metric on the set $\mathcal{U}_p$*

Proof: We shall prove the following, let $\mu_1, \mu_2, \mu_3 \in \mathcal{U}_p$

1. $W_p(\mu_1, \mu_2) = 0 \iff \mu_1 = \mu_2$,
2. $W_p(\mu_1, \mu_3) \leq W_p(\mu_1, \mu_2) + W_p(\mu_2, \mu_3)$.

It is clear that W_p is symmetric, non negative and it is finite since we are considering the space $\mathcal{U}_p$.

1. It should be clear that if $\mu_1 = \mu_2$ then $W_p(\mu_1, \mu_2) = 0$. Assume now that $W_p(\mu_1, \mu_2) = 0$ for any given $\mu_1, \mu_2 \in \mathcal{U}_p$ and let π be a probability measure such that the Wasserstein metric attains its infimum.

Notice first that π is supported on the diagonal this is, the diagonal $D = \{y = x\}$ is the biggest closed set such that $\pi(\mathbb{R}^2 - D) = 0$.

$$0 \leq \int_{\mathbb{R}^2 - D} |x - y|^p d\pi \leq \int_{\mathbb{R}^2} |x - y|^p d\pi = 0,$$

But $|x - y|^p > 0$ on $\mathbb{R}^2 - D$ and so $\pi(\mathbb{R}^2 - D) = 0$ as claimed.

Now for any $\varphi \in \mathcal{C}_b(\mathbb{R}, \mathbb{R})$ we have that

$$\begin{aligned} \int_{\mathbb{R}} \varphi(x) d\mu_1(x) &= \int_{\mathbb{R}^2} \varphi(x) d\pi(x, y) = \int_D \varphi(x) d\pi(x, y) \\ &= \int_D \varphi(y) d\pi(x, y) = \int_{\mathbb{R}^2} \varphi(y) d\pi(x, y) \int_{\mathbb{R}} \varphi(y) d\mu_2(y) \end{aligned}$$

And so we get that $\mu_1 = \mu_2$.

2. The proof of the triangle inequality is a little bit more trickier, for it needs a lemma whose proof can be found in [13].

Lemma 2.1.4 *Let μ_1, μ_2, μ_3 be probability measures on $\mathbb{R}$ and let $\pi_{12} \in \Pi(\mu_1, \mu_2)$, $\pi_{23} \in \Pi(\mu_2, \mu_3)$. Then there exists a probability measure π on $\mathbb{R}^3$ with marginals π_{12} and π_{23}*

With this lemma at hand we proceed to prove the triangle inequality.

Given μ_1, μ_2 and μ_3 choose $\pi_{12} \in \Pi(\mu_1, \mu_2)$ and $\pi_{23} \in \Pi(\mu_2, \mu_3)$. Let X_i be the support of the measure μ_i . By means of the preceding lemma, we know there exists a probability measure π with respective marginals π_{12}, π_{23} . Now we simply check that the triangle inequality holds

$$\begin{aligned}
W_p(\mu_1, \mu_3) &\leq \left(\int_{X_1 \times X_3} |x - z|^p d\pi_{13}(x, z) \right)^{1/p} \\
&= \left(\int_{X_1 \times X_2 \times X_3} |x - z|^p d\pi(x, y, z) \right)^{1/p} \\
&\leq \left(\int_{X_1 \times X_2 \times X_3} (|x - y| + |y - z|)^p d\pi(x, y, z) \right)^{1/p} \\
&\leq \left(\int_{X_1 \times X_2 \times X_3} |x - y|^p d\pi(x, y, z) \right)^{1/p} \\
&\quad + \left(\int_{X_1 \times X_2 \times X_3} |y - z|^p d\pi(x, y, z) \right)^{1/p} \\
&= \left(\int_{X_1 \times X_2} |x - y|^p d\pi_{12}(x, y) \right)^{1/p} + \left(\int_{X_2 \times X_3} |y - z|^p d\pi_{23}(y, z) \right)^{1/p} \\
&\leq W_p(\mu_1, \mu_2) + W_p(\mu_2, \mu_3).
\end{aligned}$$

■

A useful characterization about the convergence in W_p is given in the following

Theorem 2.1.1 *The following statements are equivalent, for μ_n and μ measures in $\mathcal{U}_p$ and $p \geq 1$*

1. $W_p(\mu_n, \mu)$ converges to 0
2. $\mu_n \rightharpoonup \mu$ and $\sup_n \int_{|x| \geq R} |x|^p d\mu_n$ tends to 0 as R goes to ∞ .
3. $\mu_n \rightharpoonup \mu$ and

$$\lim_{n \rightarrow \infty} \int |x|^p d\mu_n \rightarrow \int |x|^p d\mu$$

4. For every real valued continuous function φ which satisfies $|\varphi(x)| \leq C|1 + |x|^p|$ for some $C \in \mathbb{R}$ then

$$\lim_{n \rightarrow \infty} \int \varphi d\mu_n = \int \varphi d\mu$$

Proof: The proof can be found in [13].

■

In general we have the following

Lemma 2.1.5 *Let $p \geq 1$, μ_n, μ, ν_n and ν probability measures on $\mathbb{R}$ such that $\mu_n \rightharpoonup \mu$ and $\nu_n \rightharpoonup \nu$, then*

$$W_p(\mu, \nu) \leq \liminf_n W_p(\mu_n, \nu_n)$$

Proof: This proof makes use of the *Skorohod's theorem*, see [2], which states that the weak convergence of probability measures

$$\mu_n \rightharpoonup \mu$$

implies the convergence almost everywhere of the generalized inverses of the cumulative distribution functions

$$F_{\mu_n}^{-1}(w) \rightarrow F_{\mu}^{-1}(w) \text{ for almost all } w \in (0, 1).$$

With this theorem at hand we proceed as follows:

First of all for simplicity let

$$F_{\mu_n}^{-1}(w) = u_n^{-1}(w), \quad F_{\mu}^{-1}(w) = u^{-1}(w), \quad F_{\nu_n}^{-1}(w) = v_n^{-1}(w), \quad F_{\nu}^{-1}(w) = v^{-1}(w),$$

using this convention the Wasserstein metric becomes

$$W_p(\mu_n, \nu_n) = \left(\int_0^1 |u_n^{-1}(w) - v_n^{-1}(w)|^p dw \right)^{1/p}.$$

Now we get

$$\begin{aligned} \int_0^1 |u^{-1}(w) - v^{-1}(w)|^p dw &= \int_0^1 \lim_{n \rightarrow \infty} |u_n^{-1}(w) - v_n^{-1}(w)|^p dw \\ &\leq \liminf_{n \rightarrow \infty} \int_0^1 |u_n^{-1}(w) - v_n^{-1}(w)|^p dw, \end{aligned}$$

by *Fatou's lemma*, and thus

$$W_p(\mu, \nu) \leq \liminf_{n \rightarrow \infty} W_p(\mu_n, \nu_n),$$

as claimed. ■

Another useful property of the Wasserstein metric is that of convexity given by the following proposition.

Lemma 2.1.6 *Let $\mu_1, \mu_2, \nu_1, \nu_2$ be probability measures on $\mathbb{R}$, $p \geq 1$ and $\alpha \in [0, 1]$, then*

$$W_p^p(\alpha\mu_1 + (1 - \alpha)\mu_2, \alpha\nu_1 + (1 - \alpha)\nu_2) \leq \alpha W_p^p(\mu_1, \nu_1) + (1 - \alpha) W_p^p(\mu_2, \nu_2).$$

Proof: See [5] or [13]. ■

CHAPTER 3

Scalar conservation laws

3.1 Scalar conservation laws.

3.1.1 The Burger's equation

Consider the following Cauchy problem related to the partial differential equation, known as the *Burger's equation*:

$$\begin{aligned}u_t + \left(\frac{u^2}{2}\right)_x &= 0, \\u(0, x) &= u^0(x).\end{aligned}$$

By a classical solution of this equation, we mean a differentiable function

$$u : \mathbb{R}^2 \rightarrow \mathbb{R}$$

such that

$$\begin{aligned}\frac{\partial u}{\partial t}(t, x) + u(t, x) \frac{\partial u}{\partial x}(t, x) &= 0 \\u(0, x) &= u^0(x),\end{aligned}$$

for all $(t, x) \in \mathbb{R}^2$ and $x \in \mathbb{R}$, where $u^0(x)$ is a preassigned function.

The characteristic curves are given by the following ordinary differential equation

$$\frac{dx}{dt}(t) = u(t, x(t)).$$

Now, suppose there exists $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ a classical solution of the Burger's equation. Along this characteristic curve we have

$$\frac{du}{dt}(t) = \frac{\partial u}{\partial t}(t, x(t)) + \frac{\partial u}{\partial x}(t, x(t)) \frac{dx}{dt}(t, x(t)) = u_t(t, x(t)) + u(t, x(t)) u_x(t, x(t)) = 0.$$

Hence along each characteristic curve, u is constant. On the other hand we have that

$$\frac{dx}{dt}(t) = u(t, x(t)),$$

but since $u(t, x(t))$ is constant we must have that such characteristic curves are straight lines and thus the curve passing through the point $(0, x_0)$ is of the form

$$x = x_0 + u_0(x_0)t$$

and all along such straight line $u(t, x) = u_0(x_0)$.

In order to settle ideas down, suppose that the initial data u_0 is of the form

$$u_0(x) = \begin{cases} 1 & \text{if } x < 0, \\ g(x) & \text{if } x \text{ any nonincreasing function on } [0, 1] \text{ with } g(0) = 1 \text{ and } g(1) = 0, \\ 0 & \text{if } x > 1. \end{cases}$$

According to the analysis above we have that

1. for $x_0 = 0$ the characteristic line is given by $x = 0 + 1 \cdot t = t$ and hence $u(1, 1) = u_0(0) = 1$,
2. for $x_0 = 1$ the characteristic line is given by $x = 1 + 0 \cdot t = 1$ and hence $u(1, 1) = u_0(1) = 0$.

Thus the value of u at $(1, 1)$ is not even well define, so there can not be classical solution for the Burger's equation.

For a detailed explanation about the details of the so-called characteristics method, we refer to [8] or [12].

3.1.2 Weak solutions.

We have just noticed that, for the Burger's equation, there are not classical solutions, yet we have the following:

Let $\varphi \in \mathcal{C}_c^\infty([0, \infty) \times \mathbb{R})$, by multiplying on both sides of 3.1.1 yields

$$u_t \varphi + \left(\frac{u^2}{2}\right)_x \varphi = 0.$$

Integrating by parts we obtain the following formula

$$\int_{\mathbb{R}^2} u_t \varphi + \left(\frac{u^2}{2}\right)_x \varphi dx dt = \int_{\mathbb{R}^2} u \varphi_t + \frac{u^2}{2} \varphi_x = 0,$$

where it should be noted that the left hand side involves u and all of its derivatives whereas the right hand side does not. Based on this fact we state the following definition

Definition 3.1.1 Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a measurable function. We say that u is a weak solution for the Burger's equation if for all $\varphi \in \mathcal{C}_c^\infty([0, \infty) \times \mathbb{R})$ the following formula holds

$$\int_{\mathbb{R}^2} u \varphi_t + \frac{u^2}{2} \varphi_x = 0.$$

For a fine introductory exposition of the theory of partial differential equations and weak solutions we refer to [10]. We only make a couple of comments. In many cases, this notion of weak solution happens to be adequate to formulate “well posed problems”. For a well posed problem we mean a Cauchy problem where existence, uniqueness and continuous dependence between solutions and initial data, among other properties, hold.

For the kind of equations we are concerned with, this notion of weak solution is not good enough because there are weak solutions to the Burger's equation yet it is not unique, as the following example shows.

Consider the Burger's equation with initial data the *Heaviside* function:

$$H(x) = 1_{[0, \infty)}(x).$$

We have that for each $\alpha \in [0, 1]$ the following are weak solutions of 3.1.1.

$$u_\alpha(t, x) = \begin{cases} 0 & \text{if } x < \frac{\alpha t}{2}, \\ \alpha & \text{if } \frac{\alpha t}{2} \leq x < \frac{(\alpha+1)t}{2}, \\ 1 & \text{if } x \geq \frac{(\alpha+1)t}{2}. \end{cases}$$

This example was taken from [7].

3.1.3 entropy solutions.

So far we have shown that the Burger's equation not only does not possess classical solutions but, even worse, weak solutions do not constitute an adequate setting, because there is no uniqueness. So in order to overcome such difficulty we adopt a new criteria that allow us to have existence and uniqueness of solutions. This criteria is the *entropy*.

The idea is that the entropy will allow us to choose, among the weak solutions, the one that is physically plausible. We point out that this is not the only way to do so and the reader should consult [7] for a clear exposition about the entropy condition as well as some other *admissible conditions*.

Following [7] or [8] we say that a weak solution u of the Burger's equation is an entropy solution if given any

$$E : \mathbb{R} \rightarrow \mathbb{R}$$

differentiable and *convex* function, and letting F such that

$$E'(x)x = F'(x)$$

the following inequality holds in the weak sense

$$E(u)_t + F(u)_x \leq 0,$$

this means that for every $\varphi \in \mathcal{C}_c^\infty([0, \infty) \times \mathbb{R})$, $\varphi \geq 0$ the following inequality holds

$$\int_{-\infty}^{\infty} \int_0^{\infty} (E(u)\varphi_t + F(u)\varphi_x) dt dx \geq 0.$$

In particular if we take $E(x) = \pm x$ and $F(x) = \pm \frac{x^2}{2}$ we find out that entropy solutions are weak solutions.

At the end of this article we will make a clear exposition about how to find the entropy solution to equations such as the Burger's equation and some other interesting examples.

3.1.4 Scalar conservation laws.

In the sequel we shall define the class of differential equations we are interested in, and some aspects related to existence and uniqueness of solutions.

By an scalar conservation law we mean the following partial differential equation

$$u_t + f(u)_x = 0 \quad t \geq 0 \quad x \in \mathbb{R}, \quad (1)$$

$$u(0, x) = u^0(x).$$

Where $u(t, x) \in \mathbb{R}$ is an unknown function, f is real valued Lipschitz function, known as the *flux* and the initial data $u^0(x) \in L^\infty(\mathbb{R})$.

An striking fact about this kind of equations is that, in general, there do not exist classical solutions to (1), such is the case of the *Burger's equation* for example. However, in the year of 1969, the Soviet mathematician *S. Kružkov* proved the following theorem.

Theorem 3.1.1 *For every $u^0 \in L^\infty(\mathbb{R})$, there exists a unique entropy solution u to (1) in the space $L^\infty([0, \infty)) \cap \mathcal{C}([0, \infty) L_{loc}^1(\mathbb{R}))$.*

Proof: [5]. ■

Following [8], an entropy solution is

Definition 3.1.2 *$g \in \mathcal{C}([0, \infty), L^1(\mathbb{R})) \cap L^\infty(\mathbb{R} \times (0, \infty))$ is said to be an entropy solution for (1) if the following inequalities hold:*

$$\int_{-\infty}^{\infty} \int_0^{\infty} (E(g)\varphi_t + F(g)\varphi_x) dt dx \geq 0,$$

$$\lim_{t \rightarrow 0} g(t, x) = u_0(x) \text{ in } L^1(\mathbb{R})$$

For all $\varphi \in \mathcal{C}_c^\infty([0, \infty) \times \mathbb{R})$ non negative i.e. $\varphi \in \mathcal{C}^\infty([0, \infty) \times \mathbb{R})$ has compact support and $\varphi \geq 0$. In this inequality $E : \mathbb{R} \rightarrow \mathbb{R}$ is a real valued differentiable function which is convex and $F(x)$ is defined by the relation :

$$E'(x)f'(x) = F'(x), \quad x \in \mathbb{R}.$$

For a more detailed explanation about entropy solutions for systems of conservation laws, we recommend the book of [8].

The following theorem gives an insight of the situation, for it relates the scalar conservation law (1) with the set $\mathcal{U}$. Although the proof of this fact lies at the end of this paper it is nevertheless the keystone of the forthcoming work in terms of consistency.

Theorem 3.1.2 *Given $u^0 \in \mathcal{U}$ as an initial data for (1), then the entropy solution u of (1), satisfies that $u(t, x) \in \mathcal{U}$ for all $t > 0$.*

Proof: The proof will be given as a corollary at the end of theorem 5.1.1. ■

The aim of this paper is thus to prove a theorem about a certain continuous dependence between the initial data and the respective solutions, in view of this theorem, we know that entropy solutions to (1), can be regarded in a certain way as probability measures, by doing so we can use the Wasserstein distance to measure how solutions will behave having some knowledge about the initial data beforehand.

CHAPTER 4

The classical viewpoint

4.1 The classical viewpoint

This section may seem to be a little out of context because, in here seek to illustrate the method of *characteristics* for solving the scalar conservation law, and show how the Wasserstein metric behaves in this case. The important feature about this section is that, anyhow the future work is inspired on the result and methods used in here.

Theorem 4.1.1 *Given a C^1 real-valued function f on R , let u^0 and $\bar{u}^0$ in $\mathcal{U}$ be two initial data such that the associated entropy solutions u and $\bar{u}$ to (1) are classical solutions, increasing in x for all $t \geq 0$. Then for any $t \geq 0$ and $p \geq 1$ we have (with possibly infinite values)*

$$W_p(u_x(t, x), \bar{u}(t, x)) = W_p(u_x^0(t, x), \bar{u}^0(t, x))$$

Proof: First of all, since u^0 is increasing from 0 to 1 it has a *true* inverse defined on $(0, 1)$. Let $X(0, w) : (0, 1) \rightarrow \mathbb{R}$ be its real inverse.

$$u^0(X(0, w)) = w \text{ for all } w \in (0, 1).$$

Pick any $w \in (0, 1)$ and consider the following ordinary differential equation, keep in mind w is fixed.

$$\frac{d}{dt}X(t, w) = f'(u(t, X(t, w))) \text{ with } X(0, w) = w.$$

Since f' is continuous, by a famous theorem due to *Picard*, we can ensure the existence of a solution, although it may not be unique. We denote such solution by $X(t, w)$, $t \geq 0$.

We claim now that if u is a solution of (1), then we must have $u(t, X(t, w)) = w$

for all $t \geq 0$. For this take the derivative respect to t and we get

$$\begin{aligned}
 \frac{d}{dt}u(t, X(t, w)) &= \frac{d}{dt}u(t, X(t, w)) + \frac{d}{dx}u(t, X(t, w))\frac{d}{dt}X(t, w) \\
 &= -f'(u(t, X(t, w)))\frac{d}{dx}u(t, X(t, w)) + \frac{d}{dx}u(t, X(t, w))\frac{d}{dt}X(t, w) \\
 &= -f'(u(t, X(t, w)))\frac{d}{dx}u(t, X(t, w)) + \frac{d}{dx}u(t, X(t, w))f'(u(t, X(t, w))) \\
 &= 0
 \end{aligned}$$

Hence $u(t, X(t, w))$ must be constant, but since $u(0, X(0, w)) = u^0(X(0, w)) = w$, the claim is proved. By virtue of this easy observation, it follows that

$$\frac{d}{dt}X(t, w) = f'(u(t, X(t, w))) = f'(w),$$

and so

$$X(t, w) = X(0, w) + tf'(w).$$

Finally let's take into account how to reckon the distance W_p , and notice that for each $t \geq 0$, $u(t, x)$ has as true inverse the function $X(t, w)$. $\bar{X}$ denotes the result of applying the same argument with the initial data $\bar{u}^0$.

$$\begin{aligned}
 W_p(u_x(t, x), \bar{u}_x(t, x)) &= \left(\int_0^1 |X(t, w) - \bar{X}(t, w)|^p dw \right)^{1/p} \\
 &= \left(\int_0^1 |X(0, w) - \bar{X}(0, w)|^p dw \right)^{1/p} \\
 &= W_p(u_x^0(t, x), \bar{u}_x^0(t, x))
 \end{aligned}$$

■

To recap we end up this section with a corollary on the classical setting.

Corollary 4.1.2 *Given a C^2 convex flux f and two C^1 increasing initial data u^0 and $\bar{u}^0 \in \mathcal{U}_p$ for some $p \geq 1$, the following three properties hold:*

1. *the associated entropy solutions u and $\bar{u}$ are classical solutions;*
2. *$u(t, x)$ and $\bar{u}(t, x)$ belong to $\mathcal{U}_p$ and are increasing for all $t \geq 0$;*
3. *for all $t \geq 0$, we have $W_p(u_x(t, x), \bar{u}_x(t, x)) = W_p(u_x^0(t, x), \bar{u}_x^0(t, x))$*

Proof: For a complete proof we refer to [5]. We shall only prove the fact that the conservation law preserves moments.

Let $p \geq 1$, $t \geq 0$ and recall that the inverse function of $u(t, x)$ is denoted by

$X(t, w)$.

$$\begin{aligned}
 \int_{\mathbb{R}} |x|^p du_x(t, x)(x) &= \int_0^1 |X(t, w)|^p dw \\
 &= \int_0^1 |X(0, w) + t f'(w)|^p dw \\
 &\leq 2^{p-1} \left(\int_0^1 |X(0, w)|^p dw + t^p \|f'\|_{L^\infty(\mathbb{R})}^p \right)
 \end{aligned}$$

which is finite by assumption. ■

CHAPTER 5

The main setting.

5.1 The main setting.

On the last section we used the so called method of characteristics to find a solution to the scalar conservation law (1). We used the fact that the initial data u^0 is continuous, and the differentiability of the solution u .

This is no longer the case when restricting to arbitrary functions of $\mathcal{U}$, so another approach is needed to find solutions of (1) and prove a contraction property for the Wasserstein distance.

Using the notation and techniques of the previous section we define:

Definition 5.1.1 *Let $u \in \mathcal{U}$, we denote its generalized inverse by*

$$X(0, w) = \inf\{x \in \mathbb{R} | u(x) \geq w\}.$$

According to the method of characteristics, for a given $h > 0$ the inverse of the initial data evolved into

$$X(h, w) = X(0, w) + hf'(w).$$

$X(h, w)$ can be regarded as random variable on $(0, 1)$, hence it makes sense to take its cumulative distribution function $F_{X(h, w)}$ which is turn in again a function of $\mathcal{U}$

$$T_h u(x) = F_{X(h, w)}(x) = \int_0^1 1_{\{w \in (0, 1) | X(h, w) \leq x\}}(w) dw.$$

The next proposition yields some basic properties about this distribution function.

Lemma 5.1.1 *Let $h \geq 0$, $F_{X(h, w)}$ defined as above and $p \geq 1$. Then*

1. $T_h u(x) = F_{X(h, w)}$ belongs to $\mathcal{U}_p$ if u does.
2. For any $u, \bar{u} \in \mathcal{U}$ we have $W_p((T_h u(x))_x, (T_h \bar{u}(x))_x) \leq W_p(u_x, \bar{u}_x)$

Proof:

1. By definition it suffices to check that

$$\begin{aligned}
 \int_{\mathbb{R}} |x|^p dF_{X(h,w)}(x) &= \int_0^1 |X(h,w)|^p dw \\
 &= \int_0^1 |X(0,w) + hf'(w)|^p dw \\
 &\leq 2^{p-1} \int_0^1 |X(0,w)|^p + |hf'(w)|^p dw \\
 &\leq 2^{p-1} \left(\int_0^1 |X(0,w)|^p dw + h^p \|f'\|_{L^\infty((0,1))} \right) \\
 &\leq 2^{p-1} \left(\int_{\mathbb{R}} |x|^p du_x(x) + h^p \|f'\|_{L^\infty((0,1))} \right)
 \end{aligned}$$

Which is finite by assumption on u .

2. Since both $X(h,w)$ and $\bar{X}(h,w)$ have respective laws $(T_h u(x))_x$ and $(T_h \bar{u}(x))_x$, by definition of the Wasserstein distance we get

$$\begin{aligned}
 W_p((T_h u(x))_x, (T_h \bar{u}(x))_x) &\leq \int_0^1 |X(h,w) - \bar{X}(h,w)|^p dw \\
 &= \int_0^1 |X(0,w) - \bar{X}(0,w)|^p dw = W_p(u_x, \bar{u}_x),
 \end{aligned}$$

Since $X(h,w) - \bar{X}(h,w) = X(0,w) - \bar{X}(0,w)$.

■

Lemma 5.1.2 *Given $u \in \mathcal{U}$ and $h \geq 0$ we have*

$$\|T_h u(x) - u(x)\|_{L^1(\mathbb{R})} \leq 2h \|f'\|_{L^\infty((0,1))}.$$

Proof: let $X(0,w)$ be the generalized inverse of u , we then have

$$\begin{aligned}
 |T_h u(x) - u(x)| &= \left| \int_0^1 1_{\{w \in (0,1) | X(0,w) + hf'(w) \leq x\}}(w) dw - u(x) \right| \\
 &= \left| \int_0^1 1_{\{w \in (0,1) | X(0,w) + hf'(w) \leq x\}}(w) - 1_{\{w \in (0,1) | X(0,w) \leq x\}}(w) dw \right| \\
 &= \left| \int_0^1 1_{\{w \in (0,1) | X(0,w) + hf'(w) \leq x\} \triangle \{w \in (0,1) | X(0,w) \leq x\}}(w) dw \right| \\
 &\leq \int_0^1 1_{\{w \in (0,1) | 0 < |x - X(0,w)| \leq h|f'(w)|\}}(w) dw \\
 &\leq \int_0^1 1_{\{w \in (0,1) | 0 < |x - X(0,w)| \leq h\|f'\|_{L^\infty((0,1))}\}}(w) dw \\
 &\leq u(x + h\|f'\|_{L^\infty((0,1))}) - u(x - h\|f'\|_{L^\infty((0,1))})
 \end{aligned}$$

The result now follows by integration on $\mathbb{R}$. ■

With the aid of this cumulative distribution function $(T_h u(x))_x$ we construct an approximate solution for the scalar conservation law.

Definition 5.1.2 *Let $h > 0$ and $u \in \mathcal{U}$. For any $t \geq 0$ choose $N \in \mathbb{N}$ and $0 \leq s < 1$ such that $t = (N + s)h$, we write*

$$S_h u(t, x) = (1 - s)T_h^N u(x) + sT_h^{N+1} u(x).$$

Remark 4 *The definition of $S_h u(t, x)$ shows that it will be necessary to be able to reckon iterations of the T_h operator, however there is a special, somewhat general case where this task becomes simpler.*

Assume that there exists a function $g : \mathbb{R} \rightarrow (0, 1)$ such that for each $h > 0$

$$g^{-1}(w) = X(0, w) + hf'(w),$$

for all $w \in (0, 1)$.

Under this assumptions we have that the process of taking generalized inverses and that of finding the cumulative distribution functions are inverse to each other, hence when looking for $T_h^N u(x)$ for large $N \in \mathbb{N}$, the iteration process can be skipped, for it becomes

$T_h^N u(x) =$ The cumulative distribution function of the random variable, $X(0, w) + Nh f'(w)$.

The expression S_h is well defined by the previous lemma. We now ought to prove some lemmas regarding contraction properties paving the way up to our main result, a contraction property on the Wasserstein distance when applied to the solution of (1).

Lemma 5.1.3 *Pick $h > 0$ and let S_h as above. For any $s, t \geq 0$ we get*

$$\|S_h u(t, x) - S_h u(s, x)\|_{L^1(\mathbb{R})} \leq 2|t - s| \|f'\|_{L^\infty([0,1])}.$$

Proof: First of all, choose $M, N \in \mathbb{N}$ and $\mu, \nu \in [0, 1)$ such that $t = (M + \mu)h$ and $s = (N + \nu)h$. Assume without loss of generality that $M \geq N$.

$$\begin{aligned} S_h u(t, x) - S_h u(s, x) &= (1 - \mu)T_h^M u(x) + \mu T_h^{M+1} u(x) - ((1 - \nu)T_h^N u(x) + \nu T_h^{N+1} u(x)) \\ &= T_h^M u(x) - T_h^N u(x) \\ &\quad + \mu(T_h^{M+1} u(x) - T_h^M u(x)) - \nu(T_h^{N+1} u(x) - T_h^N u(x)) \\ &= T_h^M u(x) \cdots - T_h^K u(x) + T_h^K u(x) \cdots - T_h^N u(x) + \\ &\quad + \mu(T_h^{M+1} u(x) - T_h^M u(x)) - \nu(T_h^{N+1} u(x) - T_h^N u(x)) \\ &= \sum_{K=N+1}^{M-1} T_h^{K+1} u(x) - T_h^K u(x) \\ &\quad + \mu(T_h^{M+1} u(x) - T_h^M u(x)) + (1 - \nu)(T_h^{N+1} u(x) - T_h^N u(x)) \end{aligned}$$

Notice that $t \geq s$ since $M > N$ and so

$$|t - s| = t - s = (M - N + \mu - \nu)h,$$

Finally by taking the $L^1(\mathbb{R})$ norm and applying lemma 5.2 we get

$$\begin{aligned} & \|S_h u(t, x) - S_h u(s, x)\|_1 \\ = & \left\| \sum_{K=N+1}^{M-1} T_h^{K+1} u(x) - T_h^K u(x) \right. \\ & \left. + \mu(T_h^{M+1} u(x) - T_h^M u(x)) + (1 - \nu)(T_h^{N+1} u(x) - T_h^N u(x)) \right\|_1 \\ \leq & \sum_{K=N+1}^{M-1} \|T_h^{K+1} u(x) - T_h^K u(x)\|_1 \\ & + \mu\|(T_h^{M+1} u(x) - T_h^M u(x))\|_1 \\ & + (1 - \nu)\|(T_h^{N+1} u(x) - T_h^N u(x))\|_1 \\ \leq & 2(M - N - 1 + \mu + 1 - \nu)h\|f'\|_\infty \\ = & 2|t - s|\|f'\|_\infty \end{aligned}$$

as desired. ■

Lemma 5.1.4 *Let h be any positive number and S_h as above. Then for u and $\bar{u}$ in $\mathcal{U}$, we have for any $t \geq 0$*

$$W_p((S_h u)_x(t, x), (S_h \bar{u})_x(t, x)) \leq W_p(u_x, \bar{u}_x)$$

Proof: Recall lemma 5.1.1 where the result was proved for T_h , and recall the convexity property of the Wasserstein metric in lemma 2.1.6.

Let $N \in \mathbb{N}$ and $0 \leq s < 1$ such that $t = (N + s)h$, we have that

$$\begin{aligned} W_p^p((S_h u)_x(t, x), (S_h \bar{u})_x(t, x)) &= W_p^p((1 - s)(T_h^N u)_x(x) \\ &\quad + s(T_h^{N+1} u)_x(x), (1 - s)(T_h^N \bar{u})_x(x) \\ &\quad + s(T_h^{N+1} \bar{u})_x(x)) \\ &\leq (1 - s)W_p^p((T_h^N u)_x(x), (T_h^N \bar{u})_x(x)) \\ &\quad + sW_p^p((T_h^{N+1} u)_x(x), (T_h^{N+1} \bar{u})_x(x)) \\ &\leq (1 - s)W_p^p(u_x, \bar{u}_x) + sW_p^p(u_x, \bar{u}_x) \\ &= W_p^p(u_x, \bar{u}_x). \end{aligned}$$

taking p -roots

$$W_p((S_h u)_x(t, x), (S_h \bar{u})_x(t, x)) \leq W_p(u_x, \bar{u}_x)$$

And we are done. ■

5.1.1 The central theorem.

The following theorem is the heart of the paper.

Theorem 5.1.1 *Let $u^0 \in \mathcal{U}$, then $S_h u^0(t, x)$ converges to an entropy solution of (1), as h goes to 0 in $\mathcal{C}([0, \infty), L^1_{loc}(\mathbb{R}))$.*

The proof will be done in two main steps. On the first one we shall prove that actually

$$\lim_{h \rightarrow 0} S_h u^0(t, x)$$

exists for every $t \geq 0$ and $x \in \mathbb{R}$, and then we will prove that such limit is the entropy solution of the scalar conservation law.

Lemma 5.1.5 *Under the hypothesis of theorem 5.1.1,*

$$\lim_{h \rightarrow 0} S_h u^0(t, x)$$

exists.

Proof: Let $M > 0$ and let $t \in [0, M]$ note that the set

$$\{S_h u^0(t, x) | h > 0\}$$

is relatively compact in $\mathcal{U} \subset L^1_{loc}(\mathbb{R})$ by theorem 1.1.3. Since $[0, M]$ is a compact metric space, and

$$\|S_h u^0(t, x) - S_h u^0(s, x)\|_{L^1_{loc}(\mathbb{R})} \leq 2|t - s| \|f'\|_{L^\infty([0, 1])}$$

by 5.1.3, the conditions of *Arzela-Ascoli*, see [1], are fulfilled and the result now follows. $\blacksquare$

Note that every accumulation point is actually a function in $\mathcal{U}$.

The next step on the proof is to show that any accumulation point, as $h \rightarrow 0$ satisfies the entropy condition. We will momentarily denote any of such limits $Su^0(t, x)$ and we will put, with the aim of having a simpler notation throughout the proof

$$\lim_{h \rightarrow 0} S_h u^0(t, x) = Su^0(t, x),$$

nevertheless such limit, so far may not exists, we only mean a sequence h tending to 0.

Lemma 5.1.6 *Under the hypothesis of theorem 5.1.1 and lemma 5.1.5, we have the following: For every*

$$E : \mathbb{R} \rightarrow \mathbb{R}$$

differentiable and convex function, and letting F such that

$$F'(x) = E'(x)f'(x),$$

and for every test functions

$$r : [0, \infty) \rightarrow \mathbb{R}$$

and

$$\varphi : \mathbb{R} \rightarrow \mathbb{R}$$

the following inequality holds

$$-\int_0^\infty \int_{\mathbb{R}} (E(Su^0(t, x))r'(t)\varphi(x) + F(Su^0(t, x))r(t)\varphi(x)) dx dt \leq 0.$$

Proof:

1. First of all we approach the integral with respect to t by means of the following adequate Riemann sum. For $h > 0$ and $k \in \mathbb{N}$ let

$$h \sum_{k \in \mathbb{N}} r(hk) \left(\int_{\mathbb{R}} \frac{E(S_h u^0(h(k+1), x)) - E(S_h u^0(hk, x))}{h} \varphi(x) dx - \int_{\mathbb{R}} F(S_h u^0(hk, x)) \varphi'(x) dx \right).$$

We now justify the above approximate expression.

First, recall that r has compact support as well as r' . Moreover $\text{supp}(r') \subset \text{supp}(r)$ and so the integration with respect to t is carried on a compact set. Now since $h > 0$, hk defines a partition of $\text{supp}(r)$ of mesh h and so the following holds

$$-\int_0^\infty \int_{\mathbb{R}} F(Su^0(t, x))r(t)\varphi(x) dx dt = \lim_{h \rightarrow 0} -h \sum_{k \in \mathbb{N}} \int_{\mathbb{R}} F(Su^0(hk, x))r(hk)\varphi'(x) dx, \quad (\star)$$

since the right hand side is an ordinary Riemann sum. Keep in mind that $\sum$ is actually a finite sum because $\text{supp}(r)$ is compact.

On the other hand for $a > 0$, we have just shown that

$$\lim_{a \rightarrow 0} S_a u^0(t, k) = Su^0(t, k),$$

where this limit should be taken as a limit in $\mathcal{C}([0, \infty), L_{loc}^1(\mathbb{R}))$, and so for each t we have a function $S_a u^0(t, \cdot)$ in $L_{loc}^1(\mathbb{R})$.

Using the fact that φ stands for a test function and thus it is compactly supported and regularity on the entropy flux F , we have that the following integral is finite

$$\int_{\mathbb{R}} F(S_a u^0(t, x)) \varphi'(x) dx < \infty,$$

for all $a > 0$.

Fix $t > 0$, the correspondence

$$t \longrightarrow \int_{\mathbb{R}} F(S_a u^0(t, x)) \varphi'(x) dx,$$

converges *uniformly*, as $a \rightarrow 0$, to

$$\int_{\mathbb{R}} F(Su^0(t, x)) \varphi'(x) dx,$$

and so we can pass the limit to get

$$\lim_{a \rightarrow 0} \int_0^\infty \int_{\mathbb{R}} F(S_a u^0(t, x)) r(t) \varphi'(x) dt dx = \int_0^\infty \int_{\mathbb{R}} F(Su^0(t, x)) r(t) \varphi'(x) dt dx.$$

The tricky part in this proof is not only to choose, as $h \rightarrow 0$, a partition of mesh h but also involve the approximates $S_h u^0(t, x)$ of $Su^0(t, x)$.

Thus $(\star)$ holds for we are, as $h \rightarrow 0$, taking a Riemann sum of mesh h of the approximate $S_h u^0(t, x)$, and convergence is granted by our previous discussion.

We now remain to check that

$$\begin{aligned} \lim_{h \rightarrow 0} h \sum_{k \in \mathbb{N}} r(hk) \int_{\mathbb{R}} \frac{E(S_h u^0(h(k+1), x)) - E(S_h u^0(hk, x))}{h} \varphi(x) dx \\ = - \int_0^\infty \int_{\mathbb{R}} E(Su^0(t, x)) r'(t) \varphi(x) dx dt. \end{aligned}$$

We proceed as above but we now make the following remark:

For every $x \in \mathbb{R}$, the *Abel* formula [11] for finite sums yields:

$$\begin{aligned} \sum_{k \in \mathbb{N}} r(hk) \frac{E(S_h u^0(h(k+1), x)) - E(S_h u^0(hk, x))}{h} \\ = - \sum_{k \in \mathbb{N}} E(S_h u^0(hk, x)) \frac{r(h(k+1)) - r(hk)}{h}, \end{aligned}$$

Where the right hand side tends to

$$\int_0^\infty E(Su^0(t, x)) r'(t) dt$$

as $h \rightarrow 0$.

Finally by Fubini's theorem, the result follows.

2. We proceed with a very important inequality which puts the elements of this lemma all together.

Lemma 5.1.7 *For every positive test function $\varphi(x)$, every $u^0 \in L^1_{loc}(\mathbb{R})$ and E differentiable and convex function, the following inequality holds:*

$$\frac{1}{h} \int_{\mathbb{R}} (E(T_h u^0(x)) - E(u^0)) \varphi(x) dx \leq \int_{\mathbb{R}} F(u^0(x)) \varphi'(x) dx + C h e^{Ch} \|u^0\|_{L^1_{supp(\varphi)}}.$$

where $C \leq 0$ depends only on φ and E .

The proof can be found in [6]

3. We put in all the elements developed so far, i.e. the estimate and the inequality. We consider the integral

$$- \int_0^\infty \int_{\mathbb{R}} (E(Su^0(t, x)) r'(t) \varphi(x) + F(Su^0(t, x)) r(t) \varphi(x)) dx dt,$$

which can be approach by

$$h \sum_{k \in \mathbb{N}} r(hk) \left(\int_{\mathbb{R}} \frac{E(S_h u^0(h(k+1), x)) - E(S_h u^0(hk, x))}{h} \varphi(x) dx \right. \\ \left. - \int_{\mathbb{R}} F(S_h u^0(hk, x)) \varphi'(x) dx \right),$$

and by the previous inequality this quantity is less than

$$h \sum_{k \in \mathbb{N}} r(hk) C h e^{Ch} \|T_h^k u^0(x)\|_{L^1} \leq C h^2 e^{Ch} \sum_{k \in \mathbb{N}} r(hk) (2\|f'\|_{L^\infty} h + \|u^0\|_{L^1_{loc}}) \leq 0,$$

for all h because C as given in the previous lemma is negative, and all the other involved expressions are positive and so we are done. ■

Under the hypothesis of theorem 5.1.1, we have proved existence of an entropy solution of (1), we would like to point out that we have made no use of Kružkov's theorem 3.1.1 on that. The crucial point here is that our proof has shown that every accumulation point, as $h \rightarrow 0$, satisfies the entropy condition and thus is an entropy solution of (1), at this point is that theorem 3.1.1 grants unicity and thus allows us to have the following *limit*

$$\lim_{h \rightarrow 0} S_h u^0(t, x)$$

as the unique *entropy solution* of (1).

As promised before we now derive as a corollary theorem 3.1.2.

Corollary 5.1.2 *Given $u^0 \in \mathcal{U}$ as an initial data for (1), then the entropy solution u of (1), satisfies that $u(t, x) \in \mathcal{U}$ for all $t > 0$.*

Proof: We have just shown that the unique solution to (1) is an element of $\mathcal{U}$, for every $t \geq 0$. ■

5.1.2 Convergence properties.

The following is a rather technical lemma important for the proof of the fact that there is a convergence of the S_h to the entropy solution in the Wasserstein metric.

Lemma 5.1.8 *Let S_h defined as above, $u \in \mathcal{U}_p$ and T a positive real number. Then*

$$\sup_{0 \leq h \leq T} \sup_{0 \leq t \leq T} \int_{|x| \geq R} |x|^p d(S_h u)_x(t, x) \rightarrow 0 \text{ as } R \rightarrow \infty.$$

Proof: First we must consider T_h itself and without loss of generality let $R \geq h\|f'\|_\infty$

$$\begin{aligned}
\int_{|x| \geq R} |x|^p d(T_h u)_x(x) &= \int_0^1 |u^{-1}(w) + hf'(w)|^p 1_{\{w \in (0,1) | u^{-1}(w) + hf'(w) \geq R\}}(w) dw \\
&\leq \int_0^1 (|u^{-1}(w)| + h\|f'\|_\infty)^p 1_{\{w \in (0,1) | u^{-1}(w) + hf'(w) \geq R\}}(w) dw \\
&\leq \int_{\mathbb{R}} (|x| + h\|f'\|_\infty)^p 1_{\{x \in \mathbb{R} | |x| + h\|f'\|_\infty \geq R\}}(x) du_x(x) \\
&= \int_{\mathbb{R}} |x|^p \left(1 + \frac{h\|f'\|_\infty}{|x|}\right)^p 1_{\{x \in \mathbb{R} | |x| + h\|f'\|_\infty \geq R\}}(x) du_x(x) \\
&\leq \int_{\mathbb{R}} |x|^p \left(1 + \frac{h\|f'\|_\infty}{R - h\|f'\|_\infty}\right)^p 1_{\{x \in \mathbb{R} | |x| + h\|f'\|_\infty \geq R\}}(x) du_x(x) \\
&= \left(1 + \frac{h\|f'\|_\infty}{R - h\|f'\|_\infty}\right)^p \int_{|x| \geq R - h\|f'\|_\infty} |x|^p du_x(x)
\end{aligned}$$

We now study the iteration T_h^N for some $N \in \mathbb{N}$, we put the additional assumption that $R \geq Nh\|f'\|_\infty$

$$\int_{|x| \geq R} |x|^p d(T_h^N u)_x(x) \leq \prod_{j=1}^N \left(1 + \frac{h\|f'\|_\infty}{R - jh\|f'\|_\infty}\right)^p \int_{|x| \geq R - Nh\|f'\|_\infty} |x|^p du_x(x)$$

Now notice that

$$\prod_{j=1}^N \left(1 + \frac{h\|f'\|_\infty}{R - jh\|f'\|_\infty}\right) \leq \left(1 + \frac{h\|f'\|_\infty}{R - Nh\|f'\|_\infty}\right)^N \leq \exp\left(\frac{Nh\|f'\|_\infty}{R - Nh\|f'\|_\infty}\right).$$

Thus

$$\begin{aligned}
&\int_{|x| \geq R} |x|^p d(S_h u)_x(Nh, x) \\
&= \int_{|x| \geq R} |x|^p d(T_h^N u)_x(x) \leq \exp\left(\frac{pT\|f'\|_\infty}{R - T\|f'\|_\infty}\right) \int_{|x| \geq R - T\|f'\|_\infty} |x|^p du_x(x),
\end{aligned}$$

for any N and h such that $Nh \leq T$.

For the general case choose $N \in \mathbb{N}$ and $0 \leq s < 1$ such that $t = (N + s)h$ for $t, s \leq T$

$$\begin{aligned}
\int_{|x| \geq R} d(S_h u)_x(t, x) &= (1 - s) \int_{|x| \geq R} |x|^p d(T_h^N u)_x(x) + s \int_{|x| \geq R} |x|^p d(T_h^{N+1} u)_x(x) \\
&\leq (1 - s) \exp\left(\frac{pNh\|f'\|_\infty}{R - Nh\|f'\|_\infty}\right) \int_{|x| \geq R - Nh\|f'\|_\infty} |x|^p du_x(x) \\
&\quad + s \exp\left(\frac{p(N+1)h\|f'\|_\infty}{R - (N+1)h\|f'\|_\infty}\right) \int_{|x| \geq R - (N+1)h\|f'\|_\infty} |x|^p du_x(x) \\
&\leq (1 - s) \exp\left(\frac{p2T\|f'\|_\infty}{R - 2T\|f'\|_\infty}\right) \int_{|x| \geq R - 2T\|f'\|_\infty} |x|^p du_x(x) \\
&\quad + s \exp\left(\frac{p2T\|f'\|_\infty}{R - 2T\|f'\|_\infty}\right) \int_{|x| \geq R - 2T\|f'\|_\infty} |x|^p du_x(x) \\
&= \exp\left(\frac{p2T\|f'\|_\infty}{R - 2T\|f'\|_\infty}\right) \int_{|x| \geq R - 2T\|f'\|_\infty} |x|^p du_x(x)
\end{aligned}$$

The proof now follows because the right hand side tends to 0 as R goes to ∞ . $\blacksquare$

To end this section we have the following convergence theorem

Theorem 5.1.3 *Given an initial data $u^0 \in \mathcal{U}_p$, let u be the entropy solution to (1). Then for any $t \geq 0$*

$$\lim_{h \rightarrow 0} W_p((S_h u^0)_x(t, x), (S_h u)_x(t, x)) = 0.$$

Proof: We already know from theorem 5.1.1 that

$$\lim_{h \rightarrow 0} S_h u^0(t, x) = u(t, x) \text{ in } L^1_{loc}(\mathbb{R}),$$

hence $S_h u^0(t, x) = u(t, x)$ converges to $u(t, x)$ in the sense of distributions, i.e. for every $\varphi \in \mathcal{C}_c^\infty(\mathbb{R}, \mathbb{R})$

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}} \varphi(x) S_h u^0(t, x) dx = \int_{\mathbb{R}} \varphi(x) u(t, x) dx,$$

and so $(S_h u^0)_x(t, x)$ converges to $u_x(t, x)$ in the distributional sense. Recall that in this particular case distribution derivatives are probability measures and so we have the following relation. For every $\varphi \in \mathcal{C}_c^\infty(\mathbb{R}, \mathbb{R})$

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}} \varphi d(S_h u^0)_x(t, x) = \int_{\mathbb{R}} \varphi du_x(t, x).$$

We now ought to replace the function space $\mathcal{C}_c^\infty(\mathbb{R}, \mathbb{R})$ by $\mathcal{C}_c(\mathbb{R}, \mathbb{R})$. In other words we are going to get rid of the differentiability. For this we use the fact that for any continuous real valued function φ with compact support, there exists a compact set K and a sequence of $\varphi_n \in \mathcal{C}_c^\infty(\mathbb{R}, \mathbb{R})$ such that $\text{supp}(\varphi) \subset K$ and $\varphi_n \rightarrow \varphi$ uniformly on K .

$$\begin{aligned} \lim_{h \rightarrow 0} \int_{\mathbb{R}} \varphi d(S_h u^0)_x(t, x) &= \lim_{h \rightarrow 0} \int_{\mathbb{R}} \varphi d(S_h u^0)_x(t, x) \\ &= \lim_{h \rightarrow 0} \lim_{n \rightarrow \infty} \int_{\mathbb{R}} \varphi_n d(S_h u^0)_x(t, x) \\ &= \lim_{n \rightarrow \infty} \lim_{h \rightarrow 0} \int_{\mathbb{R}} \varphi_n d(S_h u^0)_x(t, x) \\ &= \lim_{n \rightarrow \infty} \int_{\mathbb{R}} \varphi_n du_x(t, x) \\ &= \int_{\mathbb{R}} \varphi du_x(t, x). \end{aligned}$$

Thus we have weak convergence of the probability measures.

Our previous lemma finishes the proof for it grants the convergence in the Wasserstein metric W_p , according to theorem 2.1.1. $\blacksquare$

Theorem 5.1.4 *Given a locally Lipschitz real valued function f on $\mathbb{R}$ and two initial data u^0 and $\bar{u}^0$ in $\mathcal{U}$, let u and $\bar{u}$ be the associated entropy solutions to (1). Then, for any $t \geq 0$ and $p \geq 1$, we have*

$$W_p(u_x(t, x), \bar{u}_x(t, x)) \leq W_p(u_x^0(t, x), \bar{u}_x^0(t, x))$$

Proof: The proof of this theorem now easily follows from all the hard work done so far. First of all, according to the proof of the theorem above, for any $t \geq 0$ $(S_h u^0)_x(t, x) \rightharpoonup u_x(t, x)$ in the weak sense of probability measures and so lemma 2.1.5 yields

$$W_p(u_x(t, x), \bar{u}_x(t, x)) \leq \liminf_{h \rightarrow 0} W_p((S_h u^0)_x(t, x), (S_h \bar{u}^0)_x(t, x)),$$

but for each $h \geq 0$

$$W_p((S_h u^0)_x(t, x), (S_h \bar{u}^0)_x(t, x)) \leq W_p(u_x^0, \bar{u}_x^0)$$

by lemma 5.1.4 and this completes the proof. ■

Finally some consequences of all the work done so far

Corollary 5.1.5 *Let f be any locally Lipschitz function on $\mathbb{R}$, $p \geq 1$ and $u^0 \in \mathcal{U}_p$. Then the entropy solution u to (1) belongs to $\mathcal{C}([0, \infty), \mathcal{U}_p)$.*

Proof: We have already shown that for every $t \geq 0$ the solution to the scalar conservation law $u(t, x)$ belongs to $\mathcal{U}_p$. Furthermore by a theorem proved by S. Kruřkov, see theorem 3.1.1, we can assert that for $0 < t < s$ $u(t, x) \rightarrow u(s, x)$ in $L^1_{loc}(\mathbb{R})$, and so following the proof of theorem above we get that

$$u_x(t, x) \rightharpoonup u_x(s, x)$$

hence it remains only to check the convergence of the moments of order p .

Let $T \geq s$. We will prove that

$$\lim_{R \rightarrow \infty} \sup_{0 \leq t \leq T} \int_{|x| \geq R} |x|^p du_x(t, x) = 0.$$

Let $\varepsilon > 0$ and R such that

$$\sup_{0 \leq h \leq T} \sup_{0 \leq t \leq T} \int_{|x| \geq R} |x|^p d(S_h u^0)_x(t, x) < \varepsilon,$$

Now choose $\phi \in \mathcal{C}_c^\infty(\mathbb{R}, \mathbb{R})$ such that $\phi(x) = 0$ on $|x| \leq R$ and $0 \leq \phi \leq 1$. On one hand we get

$$\int_{\mathbb{R}} \phi(x) |x|^p d(S_h u^0)_x(t, x) \rightarrow \int_{\mathbb{R}} \phi(x) |x|^p du_x(t, x)$$

Since $\phi(x) |x|^p \in \mathcal{C}_c^\infty(\mathbb{R}, \mathbb{R})$ and we have already shown that $(S_h u^0)_x(t, x)$ converges to $u_x(t, x)$ in the distribution sense as h goes to 0. But on the other hand we have that

$$\int_{\mathbb{R}} \phi(x) |x|^p d(S_h u^0)_x(t, x) < \varepsilon$$

for all $0 \leq h, t \leq T$ and so the limit

$$\int_{\mathbb{R}} \phi(x) |x|^p du_x(t, x) < \varepsilon$$

for all $0 \leq t \leq T$. From this we get

$$\sup_{0 \leq t \leq T} \int_{|x| \geq R} |x|^p du_x(t, x) < \varepsilon$$

proving that

$$\lim_{R \rightarrow \infty} \sup_{0 \leq t \leq T} \int_{|x| \geq R} |x|^p du_x(t, x) = 0$$

as desired. ■

For the case $p = 1$ we get a sharp result

Corollary 5.1.6 *Let f be any locally Lipschitz function on $\mathbb{R}$ and $u^0 \in \mathcal{U}_1$. Then the entropy solution u to (1) satisfies the following*

$$\|u(t, x) - u(s, x)\|_{L^1(\mathbb{R})} \leq 2|t - s| \|f'\|_{L^\infty((0,1))}.$$

Proof: By virtue of theorem 5.1.1 we know that $S_h u^0(t, x) \rightarrow u_x(t, x)$ in $L_1^{loc}(\mathbb{R})$, thus we have for all $s, t, n \geq 0$

$$\|u(t, x) - u(s, x)\|_{L^1([-n, n])} = \lim_{h \rightarrow 0} \|S_h u^0(t, x) - S_h u^0(s, x)\|_{L^1([-n, n])}$$

but for all $h > 0$

$$\|S_h u^0(t, x) - S_h u^0(s, x)\|_{L^1([-n, n])} \leq \|S_h u^0(t, x) - S_h u^0(s, x)\|_{L^1(\mathbb{R})} \leq 2|t - s| \|f'\|_\infty$$

and so

$$\|u(t, x) - u(s, x)\|_{L^1([-n, n])} \leq 2|t - s| \|f'\|_\infty.$$

The result now follows since n was arbitrary. ■

CHAPTER 6

The Burger's Equation Revisited and some other examples.

6.1 The Burger's Equation Revisited and some other examples.

We are now going to perform explicit computations illustrating how to use the theory developed so far in order to reckon out the entropy solution of the Burger's equation, with initial data

$$\begin{aligned}u_t + \left(\frac{u^2}{2}\right)_x &= 0 \\ u^0(x) &= 1_{[0,\infty)}(x).\end{aligned}$$

From theorem 5.1.1 it follows that we just need to reckon the following limit

$$\lim_{h \rightarrow 0} S_h u^0(t, x),$$

thus we proceed to calculate the iterations of the operator T_h first.

1. Recall we are using $X(0, w)$ to denote the generalized inverse of the Heaviside function u^0 .

$$X(0, w) = \inf\{x \in \mathbb{R} \mid u^0(x) \geq w\} = 0, \text{ for all } w \in (0, 1),$$

thus the generalized inverse of u^0 is the constant function 0.

2. We now let the function $X(0, w)$ to evolve along the conservation law by letting

$$X(h, w) = X(0, w) + hw = hw, \text{ for all } w \in (0, 1).$$

3. By definition of T_h we get

$$T_h u^0(x) = \int_0^1 1_{\{w \in (0,1) \mid X(h,w) \leq x\}}(w) dw = \begin{cases} 0 & \text{if } x \leq 0, \\ \frac{x}{h} & \text{if } 0 < x < h, \\ 1 & \text{if } x \geq h. \end{cases}$$

4. For $N \in \mathbb{N}$ we repeat the previous three steps N times, to find that

$$T_h^N u^0(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ \frac{x}{Nh} & \text{if } 0 < x < Nh, \\ 1 & \text{if } x \geq Nh. \end{cases}$$

With this simple calculation at hand we now proceed to reckon $S_h u^0(t, x)$.

Let $h > 0$ and choose $N \in \mathbb{N}$ and $0 \leq s < 1$ such that $t = (N + s)h$, now by definition we get that

$$S_h u^0(t, x) = (1 - s)T_h^N u^0(x) + sT_h^{N+1} u^0(x).$$

Although we can achieve from this formula an approximate solution for the entropy solution of 3.1.1, we opt to choose another way that actually will lead us to the solution in any time $t > 0$, by means of less cumbersome formulas

Let $t > 0$, we choose an adequate sequence $h_k \rightarrow 0$; namely let $h_k = \frac{t}{k}$ for all $k \in \mathbb{N}$. These adequate parameters will make $S_h u^0$ simpler and easier to hand out.

In view of theorem 5.1.1, we are allowed to do so since we have shown that the following limit exists

$$\lim_{h \rightarrow 0} S_h u^0,$$

and so does not matter how we approach to it.

A simple computation yields

$$S_{h_k} u^0(t, x) = T_{\frac{t}{k}}^k u^0(x) = \begin{cases} 0 & \text{for } x \leq 0, \\ \frac{x}{t} & \text{for } 0 < x < t, \\ 1 & \text{for } x \geq t, \end{cases}$$

which is the entropy solution at time t , since this expression does not depend on k .

Consider now this initial data

$$v^0(x) = \begin{cases} 0 & \text{if } x < 0, \\ \frac{1}{2} & \text{if } 0 \leq x < \frac{1}{2}, \\ 1 & \text{if } x \geq \frac{1}{2} \end{cases}$$

Following exactly the same three steps as in the previous example, we find out that, for any $N \in \mathbb{N}$

$$T_h^N v^0 = \begin{cases} 0 & \text{if } x < 0, \\ \frac{x}{hN} & \text{if } 0 \leq x < \frac{Nh}{2}, \\ \frac{1}{2} & \text{if } \frac{Nh}{2} \leq x < \frac{Nh+1}{2}, \\ \frac{x}{hN} - \frac{1}{2hN} & \text{if } \frac{Nh+1}{2} \leq x < Nh + \frac{1}{2}, \\ 1 & \text{if } x \geq Nh + \frac{1}{2}. \end{cases}$$

Again if we want to know the entropy solution at any $t > 0$, we choose $h_k = \frac{t}{k}$ and by definition we get

$$S_{h_k} u^0(t, x) = T_{\frac{t}{k}}^k u^0(x) = \begin{cases} 0 & \text{if } x < 0, \\ \frac{x}{t} & \text{if } 0 \leq x < \frac{t}{2}, \\ \frac{1}{2} & \text{if } \frac{t}{2} \leq x < \frac{t+1}{2}, \\ \frac{x}{t} - \frac{1}{2t} & \text{if } \frac{t+1}{2} \leq x < t + \frac{1}{2}, \\ 1 & \text{if } x \geq t + \frac{1}{2}. \end{cases}$$

Since this expression does not depend on k , we have immediately that this must be the entropy solution.

Let $p \geq 1$, we will show the behaviour of the Wasserstein metric along the solutions.

First, let's study the initial data

$$W_p^p(u_x^0, v_x^0) = \int_0^1 |u^{0-1}(w) - v^{0-1}(w)|^p dw = \int_{\frac{1}{2}}^1 \left(\frac{1}{2}\right)^p dw = \left(\frac{1}{2}\right)^{p+1}.$$

and then for any $t > 0$

$$\int_{\frac{1}{2}}^1 \left| wt + \frac{1}{2} - wt \right|^p dw = \left(1 - \frac{1}{2}\right) \left(\frac{1}{2}\right)^p.$$

Therefore the Wasserstein distance remains constant for all $p \geq 1$ and all $t > 0$, as expected.

6.1.1 The linear advection equation

This equation is of the form

$$u_t + au_x = 0$$

where $a \in \mathbb{R}$. Suppose we have been given as initial data a function $u^0 \in \mathcal{U}$. In this case, our computations of T_h^N yield the following

1. $X(h, w) = X(0, w) + ha$.
2. $T_h^N u^0(x) = \int_0^1 1_{\{w \in (0,1) | X(0,w) \leq x - Nha\}}(w) dw = \int_0^1 1_{\{w \in (0,1) | w \leq u^0(x - Nha)\}}(w) dw = u^0(x - Nha)$.
3. This shows that for every $t > 0$ the solution of the advection equation is

$$u^0(x - ta),$$

solutions are just left or right translates of the the initial data, according to the sign of a .

It is very important to point out that the observations made at 4 are satisfied in these examples.

6.1.2 Absolutely continuous independent random variables.

A random variable X is said to be absolutely continuous if there exists a measurable function

$$f_X : \mathbb{R} \rightarrow \mathbb{R}$$

called a *probability density function*, such that

$$F_X(x) = \int_{-\infty}^x f_X(y) dy,$$

where the right hand side is an ordinary *Riemann* integral.

Definition 6.1.1 *Let X, Y be random variables, defined on the same probability space. X, Y are said to be independent if for every $x, y \in \mathbb{R}$*

$$\begin{aligned} & \int_0^1 1_{\{w \in (0,1) | X(w) \leq x \text{ and } Y(w) \leq y\}}(w) dw \\ &= \int_0^1 1_{\{w \in (0,1) | X(w) \leq x\}}(w) dw \cdot \int_0^1 1_{\{w \in (0,1) | Y(w) \leq y\}}(w) dw. \end{aligned}$$

Following [2], the probability density function of the random variable $X + Y$ is given by

$$f_{X+Y}(x) = f_X \star f_Y(x),$$

the $\star$ denotes the convolution product, see [10].

We will extract the entropy solution of an scalar conservation law in the case the initial data and the $hf'(w)$ can be regarded as absolutely continuous random variables and 4 is satisfied.

Let $t > 0$ and choose the adequate $h_k = \frac{t}{k}$ for $k \in \mathbb{N}$. Following the theory developed we get

1. Let f_{u^0} and g_k be the probability density functions respect to the initial data u^0 and $khf'(w)$ respectively. $T_h^k(x) = f_{u^0} \star g_k(x)$.
2. Our assumptions allow us to explicitly find the entropy solution by means of the limit

$$\lim_{k \rightarrow \infty} f_{u^0} \star g_k(x) = f_{u^0} \star G(x),$$

where

$$G(x) = \int_{-\infty}^{\frac{x}{t}} g(y) dy,$$

and g is the density of the random variable f' .

The main advantage of this approach is that it yields an explicit formula for the solution for the reasonable price of finding out an appropriate function g , task which

might not be easy, yet on the other hand convolution has a smooth effect on functions which brings us in the case of 4, make computation easier.

In general it is a difficult task to find the cumulative distribution function of the addition of two random variables, even when they are independent. That is why we have chosen the case of absolutely continuous random variables. Also in many different fields of knowledge, such as statistics, this is precisely the situation.

An interesting remark is that we can always suppose that the initial data and the random variable f' are independent, since the limit of the scheme $S_h u^0(t, x)$ is the entropy solution of the conservation law, and it is unique by 3.1.1.

Conclusions and comments.

It is remarkable how the abstract notion of left-adjoin function became so useful everywhere when performing computations. We did not mention it explicitly but the adjoin property is also used in 4, and this remark allowed us to easily make the computations of the given examples.

The Wasserstein metric, as defined in this paper, has many applications in mathematical analysis, probability theory, physics and even economics since it provides a tool for the study of mass transportation problems. See [13]. It should be pointed out that the fact that we can use left adjoin functions to calculate it, is valid only in the real case.

A similar analysis has been done for the case a *viscous conservation law*, see [5] and the *Boltzman equation* see [13].

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